

AKSHAYAA D

AI Engineer · Production RAG, Agentic AI & LLM Evaluation

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PROFESSIONAL SUMMARY

3+ years of engineering experience, currently an AI Engineer at FourKites, owning production RAG, LLM, and Agentic AI systems for enterprise supply-chain workflows end-to-end – from early architecture through production reliability – turning ambiguous, high-stakes production problems into measurable scale, reliability, and business-value gains.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Data Structures & Algorithms, Object-Oriented Programming, Low-Level Design (LLD)

LLM & GenAI: Large Language Models (LLM), Transformers, Tokenization, Prompt Engineering, Tool/Function Calling, OpenAI API, Azure OpenAI (GPT-4o), GPT-4 Vision, Fine-Tuning Concepts, Quantization Concepts, KV Cache

RAG & Vector Search: Document Ingestion, Chunking, Hybrid Search (BM25 + Semantic + Reciprocal Rank Fusion), Reranking, Embeddings, Self-Healing RAG, Qdrant, CLIP (Multimodal Embeddings)

Agentic Systems: LangChain, LangGraph, Multi-Agent Systems, Agentic Workflows, MCP (Model Context Protocol)

Backend & Cloud: FastAPI, REST APIs, Async Processing, Kafka, Redis, PostgreSQL, ClickHouse, Docker, Kubernetes, AWS/Azure, Git

Automation & Tooling: Browser/Web Automation (Playwright), Pipedream, Claude Code

Evaluation & Observability: Hybrid Retrieval + LLM-as-Judge Scoring, Agreement-Rate Benchmarking, SigNoz, Chronosphere, Langfuse

PROFESSIONAL EXPERIENCE

Software Engineer I to II (AI/ML Team) – FourKites, Chennai

Jun 2023 – Present

Internal transition to AI team, mid-2025

- Took ownership across FourKites' production agent systems – SAM (multimodal RAG extraction on LangChain/FastAPI/Kubernetes, CLIP + Qdrant retrieval, **1.39M+ workflow executions**) and ALAN (AI booking agent on Pipedream/Playwright/GPT-4 Vision) – starting from initial design, including demoing agent behavior directly to FourKites' CEO.
- Replaced SAM's hardcoded pipeline (~12% file coverage) with a format-agnostic system (21+ templates, two-stage GPT-4o on Azure OpenAI) – fixing a multi-sheet bug that dropped 90%+ of each file's data and 28,693 garbage records, while **scaling 2.3x in file volume** at 0.1–0.3% failure.
- Root-caused a 3.8x excess in duplicate shipment creation to a write-before-read idempotency race across Kafka redeliveries – after a prior investigation cleared the wrong component – and shipped a fix driving a **99.6% reduction**, classified "P1 Avoided."
- Owned reliability and automation architecture for ALAN across 3 (later 5) carrier portals, building Playwright templates and a reusable Web Automation Studio – driving a **60.7% to 8.9%** drop in failure rate and a customer-confirmed ~\$400K value estimate.
- Analyzed ambiguous, multilingual requirements – including Spanish-language email intake across 10,000+ messages – to bring a new enterprise customer live, delivering Phase 2 in 2 months; **recognized by CTOs on both sides**.

PERSONAL PROJECTS

Resume Reality Check – Resume/JD Evidence Verifier

FastAPI, RAG, RRF, Evaluation, Gemini, ONNX, BM25, Docker

- Built a hybrid-retrieval (BM25 + semantic + Reciprocal Rank Fusion) system with an LLM-as-judge scored against a fixed rubric, separating what a resume claims from what it actually proves with cited evidence.
- Found and fixed **13 real production issues** – retrieval corpus pollution, non-determinism from LLM serving-layer routing, memory-constrained deployment failure – validated against a 17-pair hand-labeled eval set across 3 different resumes.

Swarm Onboard – Multi-Agent Repo Onboarding Assistant

Python, FastAPI, LangGraph, MCP, Redis, Pydantic, LangSmith, AST Parsing, GitHub REST API

- Designed a multi-agent system that analyzes unfamiliar GitHub repositories and generates onboarding documentation, using a deterministic graph-building layer plus an LLM-based reasoning layer with cross-agent validation.
- Verified generated documentation against **4 different GitHub repositories** varying in size and structure to confirm output quality and consistency.

EDUCATION

Sri Krishna College of Engineering and Technology

Bachelor of Technology (B.Tech), Information Technology

Jul 2019 – May 2023

CGPA: 9.34/10.0, Coimbatore, India

AWARDS

5th Place, HackCovid 2021 Hackathon – built the website for a frontline-worker mental-health platform (team of 4).